Project 1

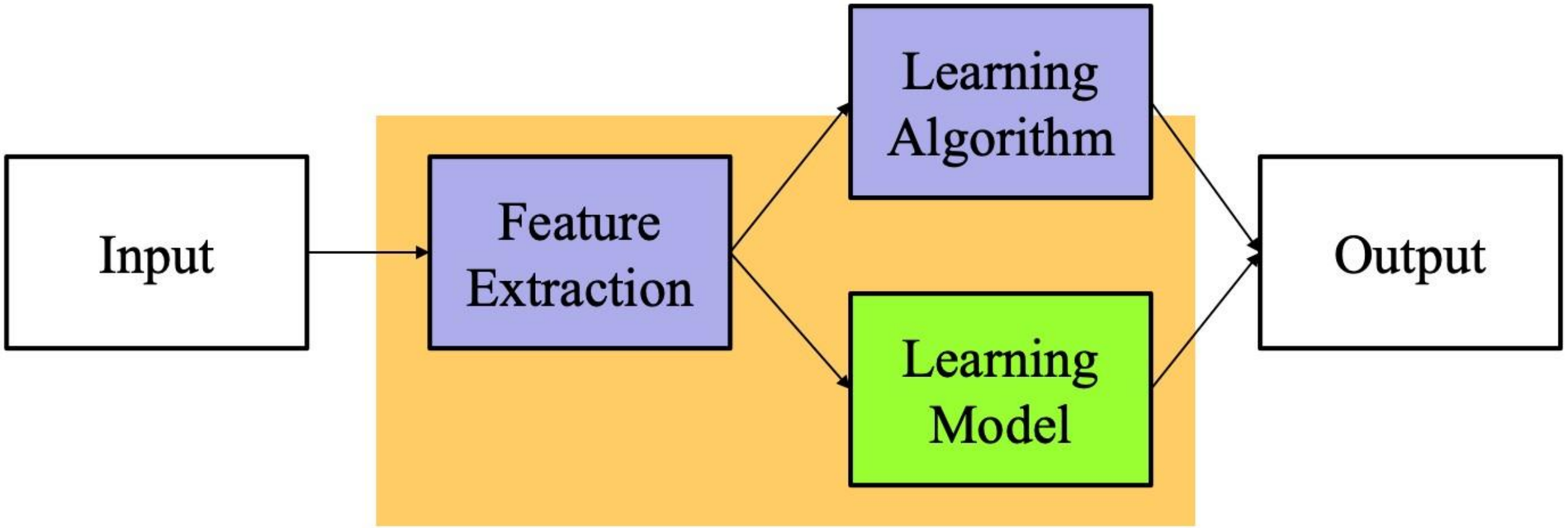
Feature Extraction:
Cats and Dogs Classification

2024 Computer Vision

Nearest Neighbor Classification

- Select three types of feature extraction (taught in class) to implement nearest neighbor.
- The three types of feature extraction can be implemented using Python's built-in functions, code references from the internet, or your own implementations.

Computer Vision & Machine Learning



: Computer Vision

: Conventional Machine Learning

: Deep Learning

Dataset

```
train/
├── Cat/
│   ├── 0.jpg
│   ├── 1.jpg
│   └── ...
└── Dog/
    ├── 0.jpg
    ├── 1.jpg
    └── ...
```

```
test/
├── Cat/
│   ├── 0.jpg
│   ├── 1.jpg
│   └── ...
└── Dog/
    ├── 0.jpg
    ├── 1.jpg
    └── ...
```





Implementation

- Data processing after feature extraction:
  - Training Set: Flatten each features of training images into a 1D vector and record the corresponding class labels.
  - Testing Set: Similarly, flatten each features of testing images into a 1D vector for distance computation.
- Distance Calculation:
  - Use SAD (Sum of Absolute Differences) as the distance metric.
  - Calculate the distance between each test image and all training images and find the training image with the smallest distance (the nearest neighbor).

Implementation

- Classification:
  - Classify the test images based on the label of its nearest neighbor.
- Accuracy Calculation:
  - Compare the predicted labels of the test set with the true labels to calculate the classification accuracy.
  - Accuracy is a key indicator of the performance of your Nearest Neighbor classification algorithm.

$$Accuracy = \frac{Number\ of\ Correct\ Classifications}{Total\ Number\ of\ Samples}$$

Note

- The assignment should be implemented by Python.

- In your report, it should contain:
  - Execution description : The commands or method you use to run the program.
  - Classification accuracy: As requirements in the slides.
  - Conclusion : The observation from your results.
  - Discussion : The questions or the difficulties you met during the implementation.

Note

- Deadline: 2024.10.30 23:59
  - Late Policy: 20% deduction per day.

- Download training & testing data from here

- Submission:
  - Zip all your files into a single one and upload it to the E-Course2
    - One python file
    - One report (only PDF)
    - File name format：studentID_project1
      ex.613410001_project1.zip